

A method to correct the optical flame transfer functions in acoustically forced, partially premixed hydrogen flame

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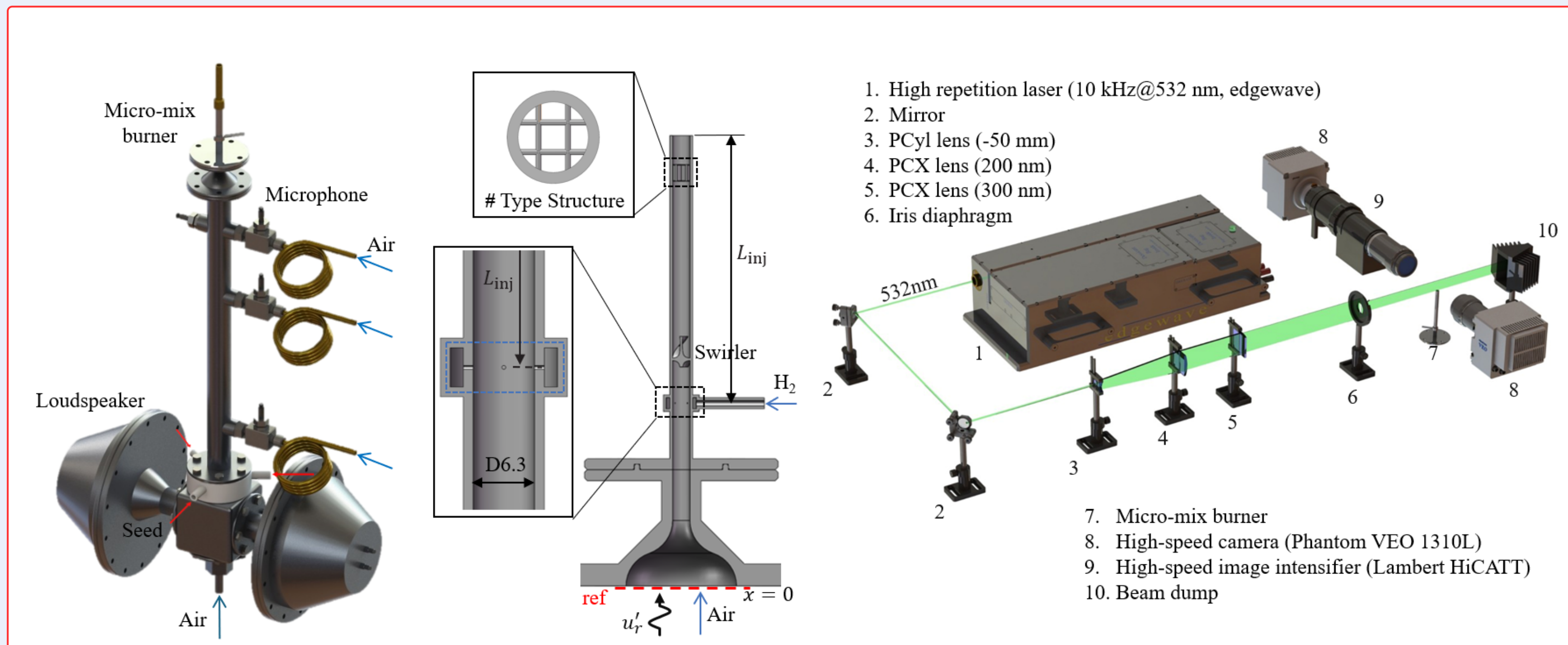
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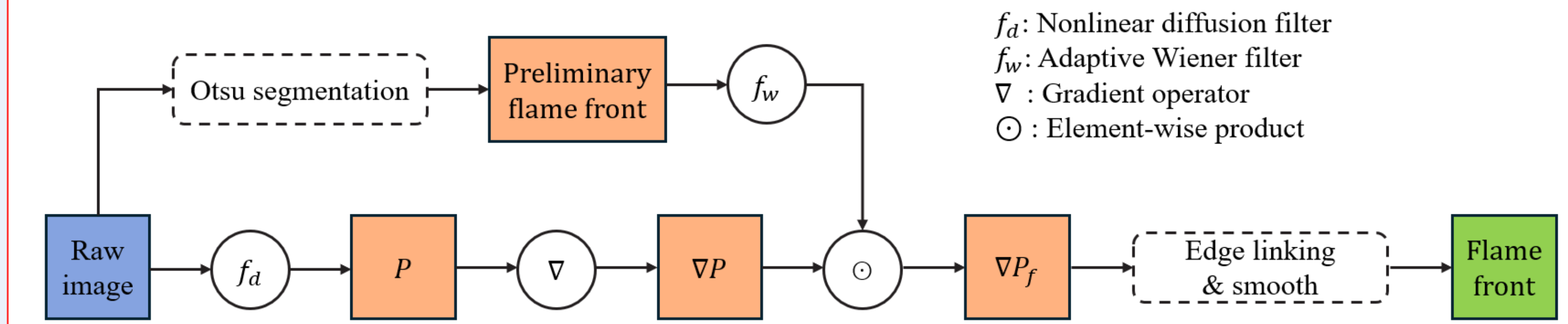
Introduction

- The response of the flame's overall unsteady heat release rate (HRR) to acoustic or equivalence ratio perturbations can be characterized by flame transfer function (FTF), which has been widely used for predicting thermoacoustic instabilities. Accurate experimental characterization of HRR is essential, yet direct measurements remain impractical because they require simultaneous determination of multiple species and temperatures.
- In practice, chemiluminescence intensity (e.g., CH* and OH* in hydrocarbon fuels) is a widely used indicator of HRR in fully premixed flames. However, this approach becomes less reliable under partially premixed conditions, where equivalence ratio fluctuations strongly influence emission intensity. In this context, developing reliable methods to construct FTFs under these conditions remains a pressing challenge.
- Motivated by the challenges discussed above, this study evaluates a correction strategy for the optical FTF, using chemiluminescence intensity and flame surface area to construct FTFs in an acoustically forced, partially premixed hydrogen flame.

Experimental Setup



Experimental setup (Liang, et al. Energy. 2026) and diagnostic system.



Flowchart of a hybrid algorithm of flame front detection (Chaib, et al., Exp. Fluids. 2023).

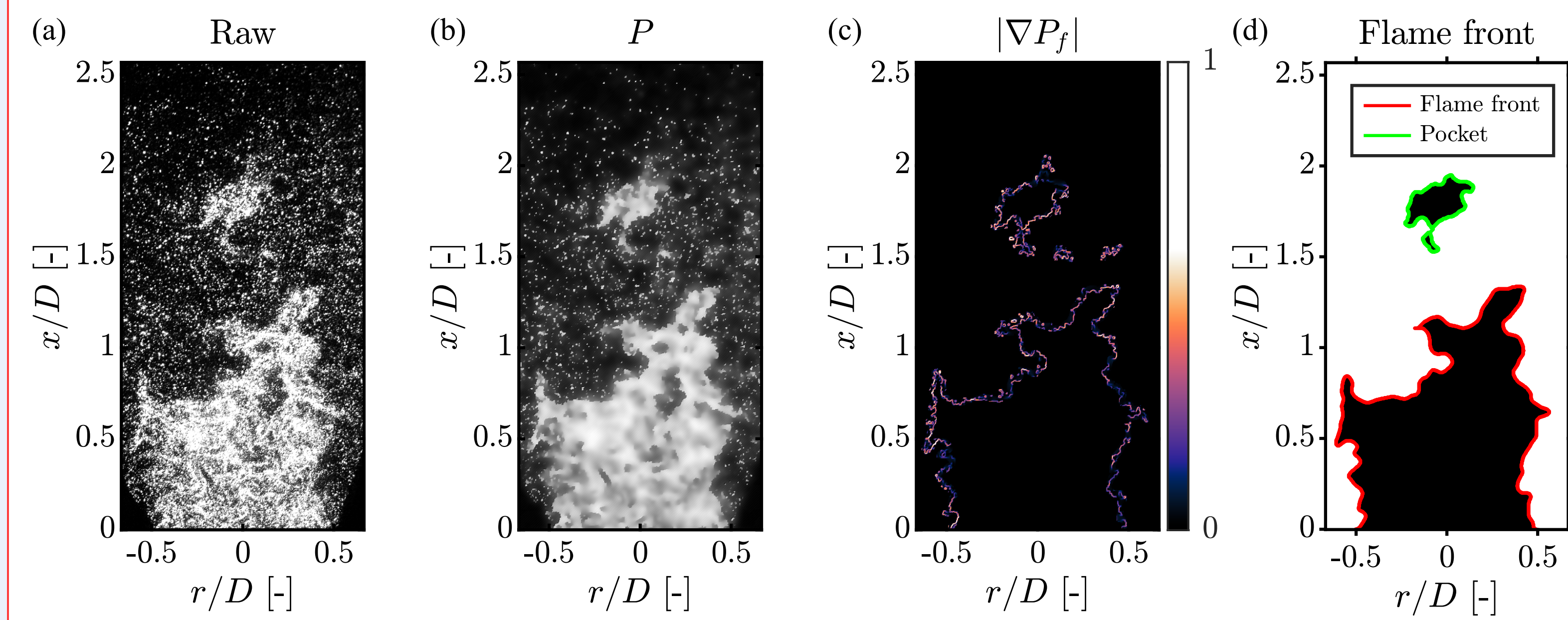


Illustration of the flame front detection process: (a) Raw Mie scattering image; (b) Pre-process image using the nonlinear diffusion filter; (c) Gradient map filtered using the binary mask; (d) Final flame front and pocket.

Key parameters:

$\bar{u}$ [m/s]	$\bar{\phi}$ [-]	$u'_{rms}/\bar{u}$ [-]	δ_L [mm]	f_s	Seed (diameter)	Stokes number
18	0.65	0.15	~0.4	10kHz	Al ₂ O ₃ (~0.1μm)	< 0.01

Two different definition of FTF:

True FTF:

$$F(\omega) = \frac{\hat{Q}(\omega)/\bar{Q}}{\hat{u}_r(\omega)/\bar{u}_r} \quad (1)$$

Optical FTF:

$$F_I(\omega) = \frac{\hat{I}(\omega)/\bar{I}}{\hat{u}_r(\omega)/\bar{u}_r} \quad (2)$$

where Q denotes the HRR and I represent the overall chemiluminescence intensity.

Summary

- The proposed methodology has been validated for laminar flames under equivalence-ratio modulation, supporting its potential applicability.
- The present work extends the method to a turbulent flame. The axisymmetric assumption may introduce uncertainty, and further quantitative validation is required.

Methodology

- Decomposition of Q' under the assumption of low-mach flow and small perturbations:

$$\frac{\hat{Q}}{\bar{Q}} \approx \frac{\hat{A}}{\bar{A}} + \frac{\int \hat{s}_L dA}{\bar{s}_L \bar{A}} + \frac{\int \hat{h}_R dA}{\bar{h}_R \bar{A}} = (F_A + F_{s_L} + F_{h_R}) \frac{\hat{u}_r}{\bar{u}_r} = F \frac{\hat{u}_r}{\bar{u}_r}, \quad (3)$$

where A denotes the flame surface area (evaluated by flame front detection), s_L and h_R is the flame propagation speed and heat of reaction, respectively.

- Linearized the relation between the overall chemiluminescence intensity, I , and the local chemiluminescence intensity, i , defined by $I = \int i dA$:

$$\frac{\hat{I}}{\bar{I}} = \frac{\hat{A}}{\bar{A}} + \frac{\int i dA}{\bar{i} \bar{A}} = (F_A + F_i) \frac{\hat{u}_r}{\bar{u}_r} = F_I \frac{\hat{u}_r}{\bar{u}_r}. \quad (4)$$

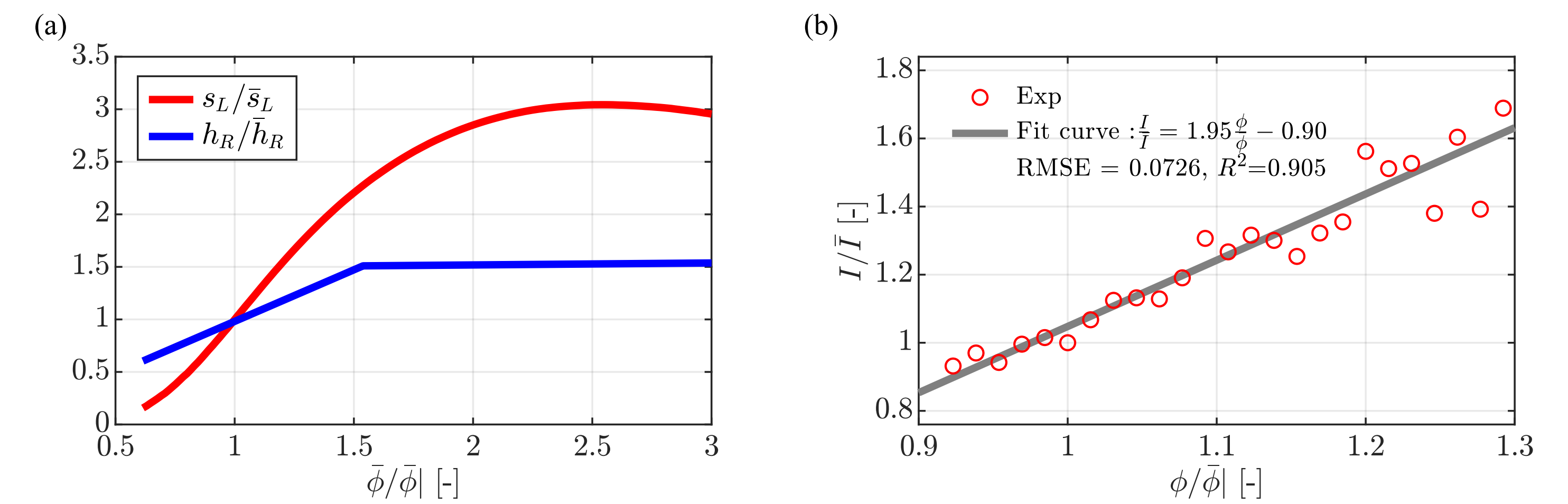
- Assuming that s_L , h_R and i only depend on the local equivalence ratio ϕ :

$$F_\phi = \frac{\int \phi dA / \bar{\phi} \bar{A}}{\hat{u}_r / \bar{u}_r} = \frac{F_{s_L}}{m} = \frac{F_{h_R}}{n} = \frac{F_i}{k}, \quad (5)$$

where $m = \left[\frac{\partial s_L / \bar{s}_L}{\partial \phi / \bar{\phi}} \right]_{\phi=\bar{\phi}}$, $n = \left[\frac{\partial h_R / \bar{h}_R}{\partial \phi / \bar{\phi}} \right]_{\phi=\bar{\phi}}$ and $k = \left[\frac{\partial I / \bar{I}}{\partial \phi / \bar{\phi}} \right]_{\phi=\bar{\phi}}$

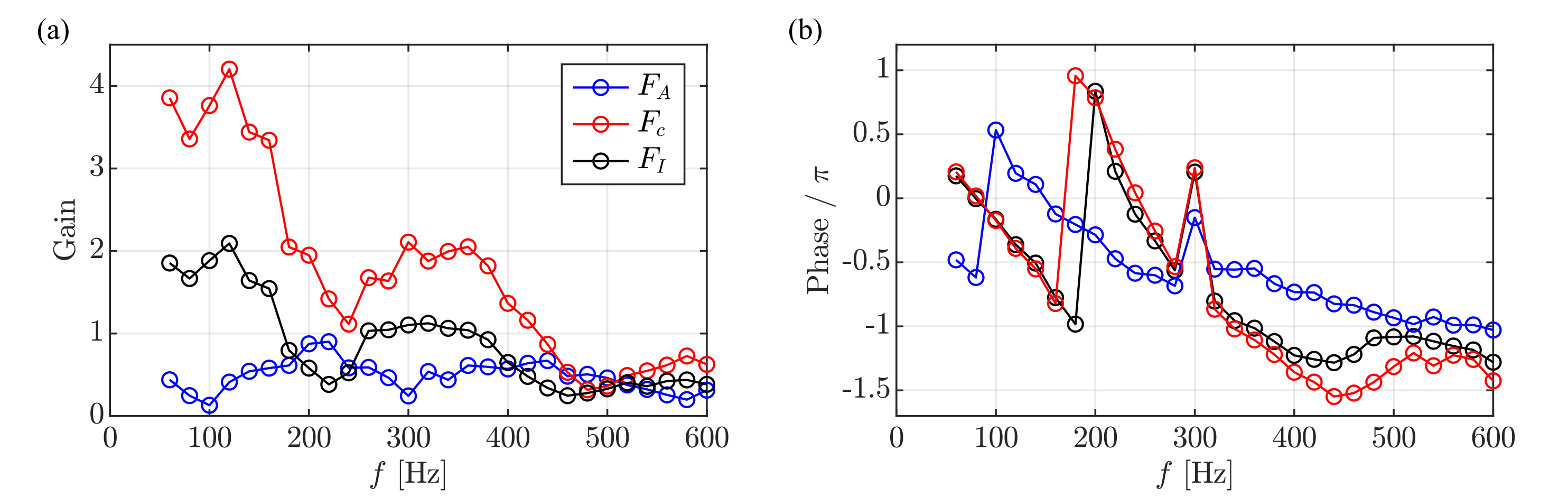
- The corrected FTF:

$$F_c = \left(1 - \frac{m+n}{k} \right) F_A + \frac{m+n}{k} F_I. \quad (6)$$



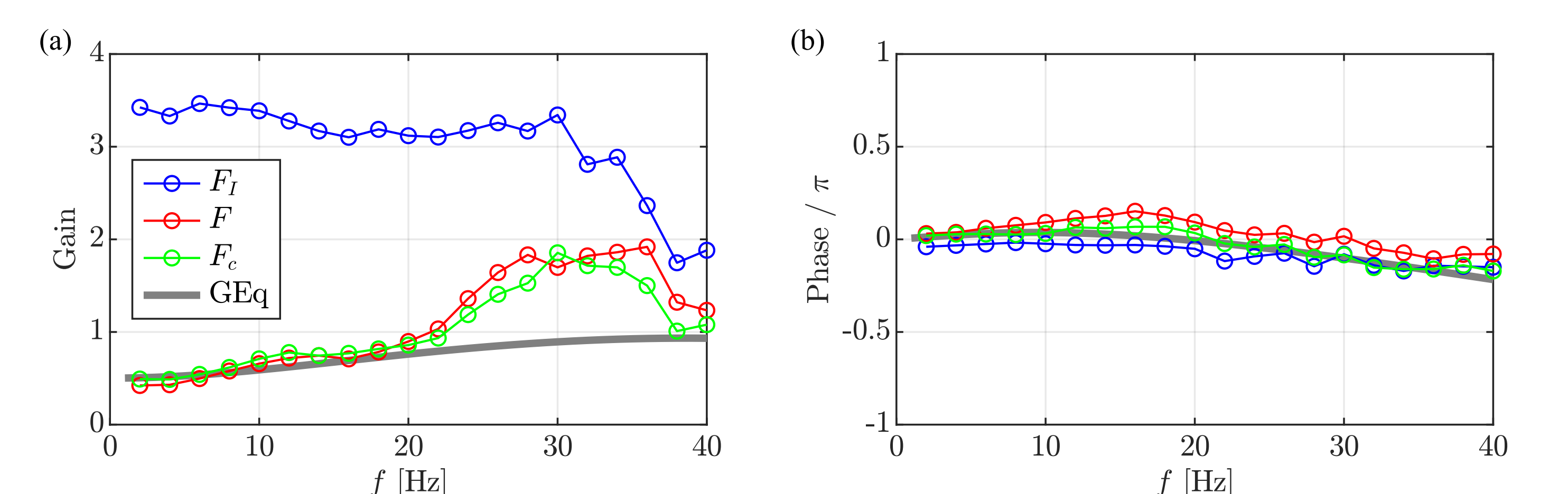
Relative variations in s_L , h_R , and I versus relative variation in ϕ at $\bar{\phi} = 0.65$.

Results



Optical and corrected FTFs (F_I and F_c), along with the measured F_A .

- Validation in a laminar conical flame ($\bar{\phi} = 1.0$):



Validation of the present method against the laminar conical flame reported in our previous work (Zhao, et al. Combust. Flame. 2025).